A Grammar-Driven Rule-Based Stemmer for Pashto

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***Abstract*—Pashto has roughly forty million speakers and very few text-processing tools. The only stemmer published for the language applies nine affix rules without a lexicon and without any means of choosing between competing analyses. We present a rule-based Pashto stemmer built from an affix inventory of 118 rules drawn from published grammars, in which every rule records its source, its productivity, and whether it may be applied at all. Candidate stems are generated separately from the decision of which candidate to keep, so a shallower analysis can displace a deeper one. No machine learning is used.**

Since no annotated set for Pashto stemming existed, we built two. A development set of 2,912 word types and a held-out set of 500 types sampled from news text were each annotated independently by three native-speaker Pashto educators, who agreed on 96.88% of stems ($\kappa = 0.969$). The held-out set was annotated with its reference column empty and was not consulted while the system was being developed. 499 of its 500 types fall within the stemming task and are the denominator below; the remaining one needs a different word and is reported apart.

On that set the proposed system reaches 72.55% exact-match accuracy against 40.48% for a faithful reimplementation of the prior system, and 72.5% against 16.9% on the types that carry an affix. Paice's under-stemming index falls from 0.889 to 0.500.

We also report a retrieval experiment, which word-type evaluation alone cannot substitute for. On 50 Pashto queries over 44,814 news documents with 1,891 relevance judgements adjudicated by three native speakers, stemming raises mean average precision from 0.5785 to 0.6854 ($p = 0.001$; 39 of 50 queries improved). One suffix is recorded in the inventory and deliberately never applied because of that experiment: removing the place-forming suffix that derives *hospital* from *healthy* merges the two, which costs retrieval more than the 0.22 points of word-type accuracy it gains. This is the only point at which the two evaluations disagree, and a word-type evaluation cannot detect it, because both analyses are defensible for a word seen in isolation.

A secondary finding concerns evaluation itself. The protocol used in the prior work, presenting system output to native speakers and asking whether it is acceptable, assigns a score of 100% to a system that modifies no word at all. We report results under both that protocol and exact match against a reference fixed in advance, and recommend the latter for Pashto stemming.

***Index Terms*—Pashto, stemming, morphological analysis, rule-based methods, low-resource languages, evaluation methodology, information retrieval**

I. Introduction

A reader searching for documents about houses does not distinguish between *house*, *houses* and *of the houses*. A retrieval system does, because to the system these are three unrelated strings. Stemming removes that difference by stripping the endings a word carries, so that related forms collapse onto a single index term. For English the problem has been settled since Porter's algorithm [2], which remains in production use after four decades.

For Pashto almost nothing exists. The language is spoken by approximately forty million people across Afghanistan and Pakistan, is one of the two official languages of Afghanistan, and has a growing volume of digital text in news and social media. It also has a rich inflectional system, which is precisely the condition under which stemming matters. Section VIII measures what that is worth: on 50 queries over 44,814 news documents, stemming recovers relevant documents an unstemmed index does not reach, though it closes only part of the gap. A classifier faces the same problem, treating the singular and plural of a key term as unrelated features and learning each from less data than it has.

The single published Pashto stemmer, by Aslamzai and Saad [1], applies nine affix rules in a fixed order with no lexicon and no selection stage. Our reimplementation of those rules leaves most words untouched, and damages a substantial fraction of the words it does modify.

Three properties of Pashto make a naive approach fail.

A. Five letters for one sound

Pashto writes five distinct characters conventionally called *ye*. They are not interchangeable; they carry gender, number and case:

Letter	Example	Function
ی	سړی	masculine singular direct
ي	سړي	masculine plural/oblique; 3rd person present
ې	ښکلې	feminine; plural
ۍ	نجلۍ	feminine singular, one noun class
ئ	ووایئ	2nd person plural imperative

Normalization pipelines written for Persian or Urdu routinely merge these into a single character. For Pashto that destroys the information a stemmer needs: سړی (man)

and سړي (men) become indistinguishable. Our system preserves all five.

B. Endings that are not endings

The same final letter may be an affix in one word and part of the root in another, with nothing in the surface form to separate the two cases: لندې reduces to لند, where ې marks agreement, while سيمه must remain سيمه, because there the ه belongs to the word. A native speaker resolves this by knowing the word; a rule conditioned on the surface form cannot.

C. Verbs whose stems cannot be derived

Pashto distinguishes weak verbs, whose stems follow from the infinitive, from strong verbs, whose present stem must be known: لیدل has the present stem وین, and تلل has خ. No affix rule derives وین from لیدل. These verbs are frequent and must be listed.

D. Contributions

- An affix inventory of 118 rules in which each entry records its grammatical source, its productivity, a confidence level, and an explicit decision about whether it may be stripped. Twenty-nine entries are documented but never applied.
- Two annotated evaluation sets for Pashto stemming, one of which was annotated blind and never used during development.
- An evaluation protocol that separates performance on word types carrying an affix from performance on types that do not, because overall accuracy alone rewards inactivity in a language where nearly half of word types are uninflected.
- A quantitative account of what each component contributes, including two components that we retain despite a measured cost and one rule that we implemented, measured and removed.
- The observation that the evaluation protocol used in prior Pashto stemming work cannot distinguish a working stemmer from an inert one.
- A retrieval test collection for Pashto: 50 queries over 44,814 documents with 1,891 relevance judgements, pooled from both runs and adjudicated by three native speakers, and the first measurement of whether Pashto stemming helps retrieval. One suffix is gated on the strength of that measurement against the word-type evidence, which is the only point where the two evaluations disagree.

II. Related Work

A. Rule-based stemming

Lovins [3] published the first English stemmer, a single-pass longest-match algorithm over 294 endings. Porter [2] replaced it with a five-step procedure whose conditions depend on a measure of syllable count. Its durability is instructive: the algorithm is small, fast, entirely inspectable, and makes no claim to produce real words. Porter maps both *relational* and *relate* to *relat*, which is not an English word. The output of a stemmer is a class label rather than a lemma, and confusing the two is a recurring error in work on morphologically rich languages.

Paice [4] contributed both an algorithm and, more consequentially for the present work, an evaluation method [5]. His under-stemming index counts related word pairs a system fails to conflate, and his over-stemming index counts unrelated pairs it wrongly merges. Because the two move in opposite directions, reporting both is more informative than accuracy alone.

B. Arabic-script languages

Khoja and Garside [6] built an Arabic root extractor that strips affixes and matches the remainder against triliteral patterns. Larkey, Ballesteros and Connell [7] showed that light stemming, removing only a short list of prefixes and suffixes, outperforms aggressive root extraction for retrieval. Their conclusion shaped the design adopted here: removing less, but removing it reliably, is preferable to removing more. For Persian, Sharifloo and Shamsfard [8] used a bottom-up parser over a morpheme lexicon. The pattern common to these systems is that a lexicon resolves ambiguity that rules alone cannot.

C. Pashto

Work on Pashto stemming is confined, to our knowledge, to Aslamzai and Saad [1]. Because that system is the baseline for everything reported here, we describe it in full.

Their algorithm applies nine rules, four removing suffixes and five removing prefixes, each with a length condition. Rule 1 removes one of eleven suffixes from words of five or more characters; Rule 2 replaces a final ي with ه in words of four or more; Rule 3 removes من; Rule 4 removes یز and appends ه; Rules 5, 6, 7 and 9 remove prefixes; and Rule 8 reduces a reduplicated word to one copy.

Reproducing the baseline exactly mattered, since a comparison against a misimplementation would be worthless. Our implementation reproduces all eight worked examples given in the paper. Four points in the publication required a decision, and we record them so that they can be checked:

- The order of prefix and suffix removal is not stated. We remove suffixes first, because the paper's own example موتروان → موتر only succeeds in that order; with prefixes first, the Rule 5 prefix م removes the first letter of موتر.
- Rule 3 is stated for words of "more than 5 characters" but its example واکمن has exactly five, so we implement the condition as five or more.
- The Rule 5 prefix list contains a bare م, which contradicts the موتروان example above. We exclude it, giving the baseline the reading its own example implies rather than the one that would lower its score.

Table I
Sources for the affix inventory.

Source	Tag
Tegey & Robson, *A Reference Grammar of Pashto* [9]	[T&R]
Robson & Tegey, "Pashto" [10]	[R&T]
David, *Descriptive Grammar of Pashto* [11]	none
Naeem & Khan, derivational morphology [13]	[N&K]
Khan, Pashto diminutives [16]	[Khan23]
Penzl, *A Grammar of Pashto* [12]	[Penzl]
Wiktionary / kaikki.org Pashto suffixes	[W]

- Rule 4 instructs the reader to remove یز and append ه. Its first example follows this, but its second gives دودیز → دود with no ه.

Rule 2 is also linguistically incorrect as published. It maps the masculine plural adjective ښکلي to ښکله, which is the feminine singular; Tegey and Robson [9] give the masculine singular ښکلی as the base form of this class. The rule is corrected in our inventory.

Finally, the reported accuracy of 87% is not comparable with the figures in this paper, and the difference lies in the protocol rather than in the algorithm. We return to this in Section VI.

D. Grammatical sources

The affix inventory was assembled from published descriptions rather than by inspecting a corpus. Table I lists the sources used and the tag each carries in the released inventory file.

Of the 118 rules, 112 cite a published source, located by chapter in Tegey and Robson [9] or by section in Penzl [12]. The remaining six (ستان, نی, نیۍ, نې, لو, ل) are entries the engine never applies. For those no grammar is being appealed to: the affix is real Pashto morphology, the reason it is recorded is that a reader would otherwise wonder why it is absent, and the reason it is not applied is measured behaviour, given with each entry in Appendix A. Every rule the engine can actually fire cites a published source.

The provenance of each rule is recorded in Appendix A rather than summarised, so a reader can check any individual entry. One distinction a single citation count would hide is worth drawing: twenty-two of the cited rules are documented and never applied. For those the citation establishes that the affix exists, not that removing it is safe, and the decision not to strip rests on measured behaviour rather than on the grammar.

III. Pashto Morphology

This section covers the parts of the morphology that bear on the design. It is not a complete grammatical description.

A. Orthography

Beyond the Arabic inventory, Pashto adds retroflex and other consonants that occur in neither Arabic nor Persian: ټ ډ ړ ږ ښ څ ځ ګ. A word containing any of these is certainly Pashto rather than a loan, and the system uses that fact.

A complication became apparent only late in this work. The same suffix is written with different *ye* letters in different registers: dictionary orthography prefers لومړی where news orthography writes لومړي. A rule spelled in one convention does not fire on the other. The system therefore compares a folded key when deciding whether a rule applies, while the letters it writes out are never altered. Section VII quantifies the effect.

B. Nominal inflection

Tegey and Robson divide Pashto nouns into masculine classes M1–M4 and feminine classes F1–F3, each with its own plural and oblique endings. The endings the system removes most often are ونه and ونو (masculine plural and its oblique), ان and انو (animate plural and oblique), ې (feminine plural) and و (oblique plural). Nominal inflection is not a simple suffix-substitution system: gender, number, noun class and the final shape of the stem can all trigger stem alternation.

C. Verbal morphology

Two productive patterns form verbs from nouns and adjectives. The بدل intransitive and ول transitive patterns are regular, so both are handled by rule and neither appears in the dictionary: جوړبدل → جوړ and جوړول → جوړ. For strong verbs the present stem cannot be predicted, and these are listed: لیدل / وین, تلل / ځ, کول / کو, غوښتل / غواړ.

D. Loanwords

Pashto has borrowed heavily from Arabic, Persian and English, and loanwords are the principal remaining source of error, because their endings frequently resemble Pashto affixes. The correct stem of منشي is منشي, not منش; of کورس is کورس, not کور, which is a different word meaning *house*; and of بېجینګ (Beijing) is the word itself, not جینګ. Arabic broken plurals are a related problem, formed by changing the internal vowel pattern rather than by adding material, so that no suffix rule can reach them.

IV. Method

Fig. 1 shows the order in which a word passes through the system. Three of the stages can return a stem on their own, and the remaining word goes to the generate-and-select stages that do the general work. Every stage can be disabled independently, which is what makes the component study in Section VII possible.

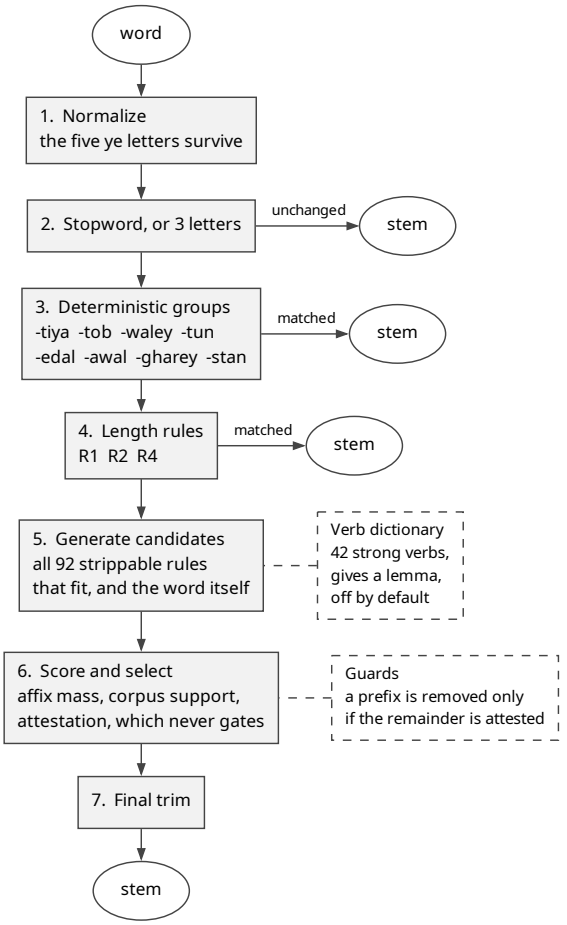


Figure 1. The pipeline. Stages 2–4 can each return a stem and stop; everything else reaches stage 5, where all applicable rules are applied and the results are then scored against one another. The two dashed boxes are not stages: the guards constrain which strips the earlier stages may make, and the irregular-verb dictionary is an optional lemmatization mode that is off by default. Affixes are transliterated in the figure for legibility and given in Pashto script in the text.

A. Normalization

Normalization removes differences that carry no information: Arabic diacritics, the tatweel elongation character, zero-width joiners, and the several Unicode encodings of the same letter. The Arabic kaf is mapped to the Pashto keheh, the alef-hamza forms to bare alef, and the Urdu letters common in Pashto written in Pakistan (ټ ډ ړ ے and the Persian گ) to their Pashto counterparts. These are encoding corrections rather than changes of spelling.

What normalization does not do is merge the *ye* letters. We verified that no *ye* character is altered in any cell of either evaluation set. This holds through the whole pipeline, not only at the normalization step: the output is always a slice of the input token, so whichever *ye* the word was written with is the one the stem carries. A rule spelled with ی that fires on a word spelled with ي returns ي. The aggressive fold remains available behind a configuration flag so that the cost of the error reported in the literature can be measured rather than asserted.

Table II
The rule schema, with three entries written out. The third is documented and never applied.

Field	ونه	توب	وکی
side	suffix	suffix	suffix
category	inflectional	derivational	derivational
min. stem	2	3	3
POS signalled	N	N	N
productivity	productive	productive	lexicalized
confidence	high	high	low
strip allowed	yes	yes	**no**
source	[T&R] M2	[N&K]	inventory §4
example	کورونه → کور	مینتوب → مین	وروکی unchanged

B. The affix inventory

The inventory holds 107 suffix rules and 10 prefix rules. Each rule records the affix and the side it attaches to; whether it is inflectional or derivational; a minimum surviving stem length; the part of speech the affix signals, where known; its productivity (productive, restricted, lexicalized or borrowed); a confidence level; and whether it may be stripped at all.

Table II gives the schema in full, with three entries written out. The last field does real work. Twenty-nine entries are documented but never applied, because applying them was measured to be harmful or because the grammatical source provides no attested example. The lexicalized diminutive وکی, the third column of Table II, is one: it survives only in fixed words such as وروکی, so removing it by rule damages them. Recording the affix and refusing to strip it is not the same as omitting it, because the inventory is also a description of the language. The minimum stem length does equally real work: without it the verb کېدل reduces to the single letter ک.

Table III groups the 118 rules by what they do. Derivation accounts for 82 of them against 36 inflectional rules, which is the opposite of what a stemmer for English would need and follows from where Pashto puts its productive morphology. The grouping also shows where the gated entries are concentrated: 17 of the 28 are noun-forming derivational suffixes, which is the area where lexicalization is commonest and where a rule is most likely to be documented in a grammar and still be unsafe to apply.

C. Length rules

Three rules were contributed by the author, a native speaker of Pashto, from his own judgement, and then tested against the annotated data. A word of three letters or fewer is returned unchanged. A word of four or five letters ending in one of the five *ye* letters or in ه loses

Table III
THE 118 RULES BY FUNCTION. "PUBLISHED" COUNTS THE RULES CITING A PUBLISHED GRAMMAR OR LEXICAL RESOURCE; "KEPT" COUNTS THOSE DOCUMENTED AND NEVER STRIPPED. THE FULL INVENTORY IS APPENDIX A.

Group	Rules	Published	Kept	Examples
Nominal inflection	23	23	0	ي انو ونه
Verbal inflection	13	11	2	ول ېدل
Derivation: nouns	57	55	18	تون تيا توب
Derivation: adjectives	14	12	4	ناک مند يز
Derivation: verbs	1	1	1	—
Prefixes	10	10	4	نيم بيا هم
Total	118	112	29	

Table IV
WEIGHTS IN THE SELECTION SCORE. THEY WERE SET BY HAND AND ADJUSTED ON THE DEVELOPMENT SET; NONE WERE FITTED AUTOMATICALLY, AND THE HELD-OUT SET WAS NOT USED TO CHOOSE ANY OF THEM.

Symbol	Meaning	Value
β	reward per character removed	0.50
F	bonus when the remainder has a word family	2.00
$A_{k>0}$	attestation bonus for a stripped candidate	2.00
A_0	attestation bonus for the unstripped word	0.50
w_f	weight on corpus frequency	0.50
γ	reward per character for an unattested strip	0.20

that letter. A word ending in ستان loses that suffix, and stripping stops there.

The last condition is necessary: allowing the general rules to continue produced ترکمنستان → ترک rather than ترکمن. The second rule originally included ل, which we removed after measuring it. On a verb ل is the infinitive ending, but on a noun it is part of the root, and the rule could not distinguish the two: it reduced کابل to کاب and لامل to لام, accounting for 51 over-strips on its own.

A fourth rule was proposed and rejected. It would have removed the last three characters of any non-Pashto word of seven characters or more. Measured against three independent references, removing a documented affix instead was more accurate in both careful annotations (45.0% against 25.0% on the development set, 51.6% against 43.5% on the held-out set). Every word the rule damaged carried a genuine affix: مسوولیت became مسوو where the affix is یت, and اعلانات became اعلا where the affix is ات. The rule fired before the inventory was consulted and cut through the root.

D. Candidate generation and selection

Generation and selection are separate, which is the central design decision borrowed from the Persian and Urdu hybrid systems. The generator applies every rule that fits and returns all distinct forms reached, including the original word with nothing removed. For کورونو the pool contains the untouched form, the result of removing ونو, and the result of removing و alone. Nothing is decided at this stage, which is what allows a shallower analysis to displace a deeper one.

A word that is already a stem is not passed through untouched by default. Below the length rules it enters the same competition as any other word, as the no-strip candidate, and it can lose: that is why the system modifies 68.9% of held-out types while the reference modifies 62.7%, and why the do-nothing floor is not a floor the system inherits.

Selection scores each candidate on how much affix material was removed, whether other words in the corpus are built on the same stem, and whether the stem itself is attested. For a candidate stem c reached by removing k characters, the score is

$$v(c) = \beta k + F\cdot[\,\mathrm{fam}(c) \geq 2\,] + \begin{cases} A_k + w_f\,\mathrm{freq}(c) & c \text{ attested} \\ \gamma k & \text{otherwise} \end{cases}$$

where $\mathrm{fam}(c)$ counts the distinct corpus words formed from c by a known suffix, $\mathrm{freq}(c)$ is the candidate's normalised corpus frequency, and A_k is the attestation bonus, which takes a smaller value when $k = 0$. Table IV gives the constants. The family term fires only when the remainder is at least three characters long.

Two of these carry the design decisions. A_0 is a quarter of $A_{k>0}$ because the unstripped word is nearly always in the corpus, and giving it the full bonus made inaction the default winner. And $\gamma > 0$ is what lets the rules act on a word the lexicon has never seen: an unattested strip still scores above zero, so out-of-vocabulary input is stemmed rather than returned whole, though it loses to any attested candidate. Two corrections to this stage were necessary during development and are worth recording, since both are easy to get wrong.

First, an early version required the stem to be an attested Pashto word before accepting a strip. That is incorrect: the output of a stemmer is a class label, and Porter's *relat* is not an English word either. Attestation now raises a candidate's score without gating it, and the system freely produces stems that are not words, including ښوون, لوبغار and پوهن.

Second, the unstripped input word is almost always present in the corpus, so awarding it the full attestation bonus made inaction the default winner and concealed every single-letter inflection. Presence in a corpus is evidence that a string is a word, not that it is a stem, and the no-strip candidate now receives a much smaller bonus.

A small number of affix groups bypass the scorer entirely; we call these the uniform-strip group, and it is the component measured under that name in Table VIII. The abstract-noun and place suffixes تيا, توب, والی and تون are long and unambiguous, so no competing analysis exists for the scorer to weigh; leaving them arbitrated meant that

روغتیا and پوهنتون remained unchanged because روغ and پوهن are absent from the lexicon.

E. Throughput

The system is a single pass of string operations over a 118-rule table, with no model to load. On one core of a commodity laptop it stems roughly 6,800 word types per second cold and 7,100 warm, measured over the development set. For an indexing pipeline the stemmer is not the bottleneck.

F. Guards

Two guards were added after error analysis. A prefix is removed only when what remains is itself an attested word. Without it the system strips لا from لانسیټ, leaving نسیټ, and from لاانتها, leaving انتها; with it, both words are returned whole. The guard is conservative in both directions, since لاانتها genuinely is لا + انتها, and we accept the missed strip in exchange for not cutting into loanwords.

The second guard concerns the interaction between the prefix rules and the length rules. A word of four or five letters that begins with a removable prefix is passed to the full inventory instead of simply losing its last letter. Only a prefix that may actually be removed counts: نا is kept rather than stripped, so ناوړه is still handled by the length rule and becomes ناور, which is what the annotation gives.

A third component was implemented and then removed. A compound splitter returned the head of a word it judged to be a compound, which conflicted with the annotation policy's decision to keep compounds whole. Measured, it fired on seven types across the two evaluation sets and was wrong on all seven, usually because the "head" it returned was itself a suffix: نړۍوال → وال and پټرولیم → یم. Disabling it is worth 0.22 points on the development set and 0.20 on held-out text. Splitting Pashto compounds reliably needs a lexicon of free forms that we do not have, and guessing at the boundary was worse than leaving it alone.

G. The verb dictionary and the scope of the task

The system includes a dictionary of 38 strong verbs covering 315 surface forms, of which 23 are checked against a printed grammar and 15 are marked unverified. The dictionary maps an inflected form to its verb: شو → کېدل. That output is usually the infinitive (182 of the 315 forms come back as the infinitive and the remaining 133 as a stem, because the ل is then removed by a suffix rule), and an infinitive is a lemma, so substituting one word for another is lemmatization rather than stemming. On the word types whose reference is a truncation, the dictionary changes the output on 69 of them and is right on 11, because it supplies the wrong kind of answer; it costs 1.36 points on the development set and 0.80 on held-out text.

The dictionary is therefore disabled by default and provided as an explicit lemmatization mode. We report its contribution separately in Section VII rather than folding it into the stemming result.

V. Data and Annotation

No annotated set for Pashto stemming was available, so two were constructed. The distinction between them determines how the results should be read.

A. Corpus

Word frequencies were taken from a corpus of Pashto news and general text assembled from VOA Pashto, BBC Pashto and a topic-labelled collection, totalling approximately 5.4 million tokens and 68,850 distinct word types. The lexicon shipped with the system is a 5,443-type frequency list drawn from this material.

Building a larger lexicon from the full corpus was attempted and made results worse. The reason is instructive: a raw corpus contains inflected forms as well as base forms, so with 48,000 types the attestation signal ceased to mean "this is a stem" and began rewarding over-stripped forms that happened to occur. The smaller list was better by being sparser.

B. Development set

2,912 word types were annotated independently by three native speakers of Pashto before the rule inventory was settled: the author, the principal of a Pashto-medium school, and a university teacher of the language. None of the three saw another's work, and none saw any system output. The author then adjudicated the disagreements and produced the released reference.

Two of the three passes are released as separate files, and the agreement between them can therefore be recomputed rather than asserted. On the 2,725 types both annotated, they chose the same stem for 96.88%, which is a Cohen's κ [14] of 0.969. Reduced to the decision that matters most, whether the word carries an affix at all, they agree on 98.24%, $\kappa = 0.965$. The two coefficients answer different questions. Chance agreement on the full stem string is 0.001, because the label space is every substring of the word, so there κ adds little to the raw figure; on the binary decision chance agreement is 0.503, and the κ is doing real work. Both fall in the band Landis and Koch [15] describe as almost perfect agreement.

The disagreements are small in substance as well as in number. Of the 86 types the two annotators stem differently, 84% differ by one or two characters: مرست against مرس, خون against خو. They are disagreements about how deep to cut, not about which analysis applies. The adjudicated reference stays close to both passes: 93.64% identical to the school principal's and 94.98% to the university teacher's. Where it departs from both, it follows the written annotation policy, and the 42 policy decisions taken during the project are logged individually with their reasons.

One part of the construction is circular and we state it rather than leave it to be found. 116 verb forms were

revised for consistency with the verb analysis, and 81 of those were changed to the reading the system itself produces, which is 3.0% of the stemming types. That is a real dependency between the reference and the system, it is confined to verbs, and it is why the held-out set rather than this one is treated as the result.

C. Held-out set

500 word types were sampled at random from the news corpus under three constraints: a frequency of at least five, to exclude typographical noise; a length of at least three characters; and no overlap with the development set. The sample was annotated by the same three native speakers, with the reference field initially empty so that nothing anchored them to any system's output. All three agreed on the result, and it is released as a single reference column. The set was not consulted during development, and every figure reported for it comes from a single run of the finished system.

The first pass was audited by comparing every removal against the affix inventory. On the released first-pass file, 73 of 327 removals, 22.3%, match no documented affix: words such as مزاج and حارث, which carry no affix at all, had been shortened. Thirty rows were revised in response, and the figure falls to 54 of 313, or 17.3%. The same audit on the development set leaves 108 of 1,370 removals, or 7.9%, unaccounted for.

That audit procedure is reusable and we recommend it. For each annotated pair, the material removed is compared against the affix inventory; a removal matching no affix is either an annotation error or evidence that the inventory is incomplete, and inspection separates the two. Applied to the development set, the same procedure identified six affixes the inventory lacked.

We went through all 54 removals that the audit leaves unaccounted for in the final held-out set, and the split is informative. Twenty-nine of them take material off one end of the word. Nineteen of those are gaps in the inventory rather than errors in the annotation, and most are variants of affixes already held: the oblique infinitive ولو beside ول, the oblique کړو beside کړ, the feminine یزې beside یزه, the past participle ېدلې beside ېدل. Two more are combinations the engine cannot form because it strips one layer at a time: توبونو is توب + ونو, and both halves are in the inventory. Eight are genuine annotation errors, and they cluster where one would expect: the English loans ایمرجینک and ریسایکلنک, where نک is not a Pashto affix, the personal name ضیالرحمان, the place name پچیراکام. Two are ambiguous.

The remaining twenty-five take material off both ends at once, and all of them are verbs: وښیي → ښي, رارسېدو → رسېد. The ending is in the inventory in every case; the prefix is a perfective or directional particle, and those are not held, for the reason Section IV.A gives: measured against both sets they were right between 13% and 50% of the time. These rows are therefore neither annotation errors nor an oversight in the inventory but the cost of that decision, stated here rather than left out of the count. An earlier version of the audit script reported only one-ended removals and so did not list them at all.

That the one-ended errors run three to one towards missing affixes rather than bad annotation is the useful part: it says the reference is sounder than the rule list, and it names the nineteen entries that would close the gap. The classified list is released with a verdict and a reason on every row.

D. Annotation policy

The policy the annotations follow is released with the data. Its main provisions are that the output is a stem and need not be a word; that Arabic broken plurals are handled by dictionary lookup rather than by rule; that proper nouns receive no special treatment, so افغانستان loses its ستان like any other word; that compounds are kept whole; and that where two analyses are both defensible, the shallower is preferred.

VI. Evaluation Protocol

A. Why exact-match accuracy alone is insufficient

The natural measure is the proportion of word types for which the system produces the annotated stem. On its own it is misleading. Nearly half of Pashto word types carry no affix and should be returned unchanged, so a system that modifies nothing scores 47.77% on the development set and 37.27% on the held-out set. Every letter a system declines to remove earns it credit.

Tuning against overall accuracy therefore drives a stemmer towards inactivity. During this work an apparent improvement of nearly three points proved, on inspection, to be exactly that: the system had become more conservative, losing seven points on the word types requiring a strip while gaining twelve on those requiring none.

All results below report three figures: accuracy on the types the annotation strips, accuracy on the types it leaves unchanged, and overall accuracy. The first cannot be obtained by inaction.

B. The scope of the stemming task

Stemming removes characters; it cannot insert them. Where the reference is an Arabic broken plural or a suppletive form (اثارو → اثر, نجونو → نجلۍ), the answer is a lemma that no affix rule can produce. Marking a truncating stemmer wrong on those types penalises it for not being a lemmatizer.

We therefore score the two tasks separately. A type belongs to the stemming task when the reference is a substring of the word, which is exactly the case where it can be reached by removing material from one end or from both; comparison is insensitive to which *ye* letter is written, since that is a difference of register and not an affix. By this criterion the development set contributes 2,717 stemming types and 195 lemma types, and the held-out set 499 and

Table V
THE SAME 499 HELD-OUT WORD TYPES UNDER BOTH PROTOCOLS. THE FINAL COLUMN GIVES THE PROPORTION OF TYPES THE SYSTEM MODIFIES AT ALL.

System	A: exact	B: judged	Modified
Modifies nothing	37.27%	**100.00%**	0.0%
Aslamzai & Saad [1]	40.48%	80.76%	37.5%
Proposed	**72.55%**	**98.60%**	68.9%

1. Of the 195, 155 are verb forms whose reference is an infinitive that no truncation reaches, شوه → کېدل and وایی → ویل; the remaining 40 are suppletive or irregular nominals such as نجونو → نجلی and Arabic broken plurals. The boundary between the two tasks is therefore almost entirely the strong verb. Results for the lemma types are reported separately in Section VII.

C. Two protocols compared

Aslamzai and Saad report 87% accuracy; our reimplementation of the same nine rules reaches 40.48% on held-out data. The difference is entirely methodological, and the point is important enough to measure rather than assert.

Protocol A, used throughout this paper, fixes the reference before any system output is seen and counts a result only if it matches exactly. Protocol B reproduces theirs: a judge is shown the output and decides whether it is an acceptable analysis. We implement Protocol B reproducibly by counting an output as acceptable when nothing was removed that is not a documented affix, and by treating an unchanged word as always acceptable, which is how a judge shown an unchanged word would score it.

The first row of Table V is the finding. Under Protocol B a system that modifies no word scores 100%, because every unchanged word is acceptable to a judge and it produces nothing else. The protocol cannot separate a working stemmer from an inert one, and a figure obtained under it cannot be compared with an exact-match figure.

Nothing here shows the published 87% to be incorrect. It shows that the measurement answers a different question. We note that the proposed system also outperforms the baseline under Protocol B, by 98.60% to 80.76%, so the conclusion does not depend on the choice.

VII. RESULTS

A. Held-out evaluation

Table VI reports both evaluation sets. The held-out set is the primary result: 499 word types from news text, annotated with the reference field initially empty and never used during development. The development set is reported beside it for comparability with the baseline, which is scored identically, and because Paice's indices are more stable on the larger set.

Table VI
BOTH EVALUATION SETS. "AFFIXED" AND "BARE" ARE ACCURACY ON THE TYPES THE ANNOTATION CHANGES AND LEAVES UNCHANGED; UI IS PAICE'S UNDER-STEMMING INDEX, WHERE LOWER IS BETTER.

System	Accuracy	Affixed	Bare	UI
Held-out set, 499 word types				
Modifies nothing	37.27%	0.0%	100.0%	1.000
Aslamzai & Saad	40.48%	16.9%	80.1%	0.889
Proposed	**72.55%**	**72.5%**	72.6%	**0.500**
Development set, 2,717 stemming types				
Modifies nothing	47.77%	0.0%	100.0%	1.000
Aslamzai & Saad	50.68%	12.2%	92.8%	0.919
Proposed	**80.97%**	**80.6%**	81.4%	**0.296**

The two systems fail in opposite directions, and that is more informative than the gap between them. Of the baseline's 297 held-out errors, 254, or 86%, are under-strips, and 163 are words it left untouched altogether; only 9% are over-strips. Of our 137, 53% are over-strips and 42% under-strips. The baseline's problem is inaction, ours is excess, which is the trade a stemmer has to make and the reason the two halves of the task are reported separately.

The proposed system improves on the baseline by 32.1 points overall. The separation of the two halves is more informative than the overall figure: on the types that require a strip the baseline reaches 16.9% and the proposed system 72.5%, a factor of four. Paice's under-stemming index falls from 0.889 to 0.500, and over-stemming remains below 4×10^{-5}.

B. Development set

The development figure is higher than the held-out one. The revisions described in Section V-B account for 0.07 points of that gap, so the explanation lies elsewhere: it is orthographic, and Section VII-E gives it. On the types requiring a strip the margin over the baseline is wider still than on held-out text: 80.6% against 12.2%, a factor of six.

C. Two external datasets

After the system was finalised, it was run against two independently annotated Pashto word lists of 5,000 and 10,000 types. Neither was produced by this project, and 96% of their entries appear in neither of our evaluation sets. Exact-match accuracy is 72.46% and 70.37% (Table VII), against 51.24% and 47.95% for the baseline. Those figures sit beside the 72.55% measured on held-out news text, which is the point: the system does not depend on the data it was built against.

Both lists carry a reference column normalised to ی while their word column retains ي, so a stem is scored wrong whenever it preserves the letter the word actually has. Folding the five *ye* letters together for the comparison alone raises the two figures to 80.32% and 77.38%. The same folded comparison is worth 1.44 points on our development set and 0.80 on our held-out set, both of which are

Table VII
TWO INDEPENDENTLY ANNOTATED WORD LISTS. THE FOLDED COLUMN COMPARES WITH THE FIVE *YE* LETTERS TREATED AS EQUAL; THE STEMMER'S OUTPUT IS NEVER REWRITTEN.

Dataset	Types	Exact	Folded
External, 5,000 types	5,000	72.46%	80.32%
Aslamzai & Saad		51.24%	57.36%
External, 10,000 types	10,000	70.37%	77.38%
Aslamzai & Saad		47.95%	53.68%
Our held-out set	499	72.55%	73.35%
Our development set	2,717	80.97%	82.41%

Table VIII
EACH COMPONENT REMOVED IN TURN. CHANGES IN PERCENTAGE POINTS.

Configuration	Development	Held-out
Full system	**80.97%**	**72.55%**
− suffix rules	71.73% (−9.24)	58.72% (−13.83)
− length rules	79.17% (−1.80)	70.34% (−2.20)
− corpus lexicon	80.27% (−0.70)	72.55% (+0.00)
− uniform-strip group	80.71% (−0.26)	72.34% (−0.20)
− prefix rules	80.93% (−0.04)	72.75% (+0.20)
+ verb dictionary	79.61% (−1.36)	71.54% (−1.00)

internally consistent in their orthography. The size of that difference is what the measurement shows: seven to eight points on the external lists against one on ours places the disagreement in how they spell the reference rather than in the stemming.

One caveat limits what these lists establish. On the words they share with our development set the two references agree 70.8% and 64.1% of the time. Sorting the disagreements shows they are a difference of policy rather than noise. About a quarter, 19.1% on the 5,000-type list and 24.3% on the larger one, are cases where our reference gives a lemma that no truncation reaches, اخيستل → اخلی against their اخ; those are not disagreements about stemming at all but about where stemming stops. Of the rest, roughly half cut deeper than we do (تنکی → تن against our تنک) and about a quarter cut less (استازیتوب → استازی against our استاز). Only two disagreements in 154 come down to the spelling of a *ye*. The two annotations differ consistently in how deep a stem should go and in whether verbs are reduced to a stem or to an infinitive, which is why these lists test generalisation rather than correctness under our policy.

D. Component study

The suffix inventory does most of the work, and removing it costs 13.8 points on held-out text. The length rules contribute between two and three points on held-out text and just under two on the development set.

Three rows record decisions taken against the measured score, and we state them explicitly rather than quietly adopting whichever setting scored best.

Prefix stripping costs a few hundredths of a point on the development set. We retain it because هم, بیا, نیم and لا are genuine Pashto prefixes, and removing the rules would amount to claiming the language has none; the attestation guard described in Section IV-F is what makes them safe. The small effect is itself explained by the inventory: of the ten prefix rules, four are documented and never stripped. نا, بې and غیر are negation prefixes that the annotation policy keeps, because ناقانونه is not an instance of قانون but its opposite, and سر is a free noun as often as it is a prefix.

The corpus lexicon is worth 0.18 points on the development set and −0.40 on held-out text. Its contribution is now marginal, because the uniform-strip and length rules decide most cases before the scorer is consulted. We report this plainly: the system should not be described as lexicon-driven.

Enabling the verb dictionary costs 1.36 points on the development set and 0.80 on held-out text, for the reason given in Section IV-G. Its proper contribution is to the lemma types, where it answers 51 of the 195 in the development set against 8 without it.

E. Orthography

The gap between the two evaluation sets has one dominant cause. Of the 500 held-out news types, 118 contain the hard *ye* ي; of the 2,912 development types, none do. The inventory contained thirteen rules ending in the narrow *ye* ى and none ending in ي, so a large class of news words passed through untouched: لومړي and تاریخي were returned unchanged.

Making rule matching insensitive to which *ye* is written, while leaving the output letters untouched, corrected this. On the development set, which is written entirely in one convention, exact matching scores 0.15 points higher; on held-out news text the folded comparison is worth 1.20 points, so it earns its place on the set that reflects real text. This is the clearest instance in the project of a defect that a single-source evaluation set could not have revealed.

VIII. RETRIEVAL EVALUATION

The evaluation so far scores the stemmer against a reference stem for each word type. That measures whether the analysis is right, not whether it helps. A stemmer exists so that a query for کور also reaches documents containing کورونه, and no word-type figure can say whether that happens or at what cost. This section reports a retrieval experiment over running text.

A. Test collection

The collection is 44,814 passages of Pashto news text, 8.0 million tokens, with a median of 182 tokens per passage, formed by grouping consecutive sentences. It was scraped from BBC Pashto, VOA Pashto, Pajhwok Afghan News, Bakhtar News Agency, Shamshad News, TOLOnews, Afghanistan International and *Nan Taki Asia*, and is a separate and larger body of text than the 5.4-million-token frequency corpus the lexicon was drawn

Table IX
RETRIEVAL OVER 44,814 DOCUMENTS, 50 QUERIES AND 1,891 RELEVANCE JUDGEMENTS. p IS TWO-TAILED; "W/L" COUNTS QUERIES IMPROVED AND HARMED. THE LOWER PANEL IS THE TWO PLACE-FORMING SUFFIXES OF SECTION VIII-D, TESTED THE SAME WAY ON THE SAME JUDGEMENTS.

Measure	No stemming	Stemmed	Δ	p	W/L
MAP	0.5785	**0.6854**	+0.1070	**0.001**	39/11
nDCG@10	0.7162	**0.7583**	+0.0420	0.226	24/13
P@10	0.6780	**0.7240**	+0.0460	0.165	20/11
Recall@10	0.3011	**0.3502**	+0.0490	**0.042**	30/11
MAP under the suffix settings tested					
Released, تون never applied		**0.6854**			
تون stripped, as the grammar says		0.6739			
ستان also gated		0.6848			
both changed		0.6733			

from, with which it shares two outlets. Fifty queries were written across the topic areas the collection covers, in the form a reader would type, between two and four content words.

Judgements were pooled: for each query the top 25 documents from each run were taken, the union was shuffled under a fixed seed, and the assessors saw the full text of each document with no indication of which run had retrieved it or at what rank. Pooling from both runs is necessary. An earlier version pooled from the unstemmed run alone, so a document only the stemmed run could find was never judged and counted irrelevant, which scores one system against a pool built to favour the other.

Each document was judged by three native Pashto speakers working together. They read the document, discussed whether it answered the query, and recorded a label only when all three agreed, so the collection carries adjudicated labels rather than one assessor's opinion or a majority vote. The result is 1,891 judgements, 1,184 of them relevant, with between 26 and 57 judged documents per query.

B. Retrieval setup

Both runs use Okapi BM25 [17] with $k_1 = 1.5$ and $b = 0.75$ in the `rank_bm25` implementation [19], retrieving 1,000 documents per query. They differ in one respect only: documents and queries pass through the stemmer in one and not in the other, with the same normalizer and tokenizer in both. Across the collection the stemmer reduces 114,644 word types to 72,173 stems, a reduction of 37.0%.

We report MAP, nDCG@10, P@10 and Recall@10, with a paired t-test and an exact sign test over the 50 per-query scores. Both are given because per-query average precision is bounded and skewed, so the t-test's normality assumption is not safe at this sample size.

C. Results

Stemming raises MAP from 0.5785 to 0.6854 (Table IX), significant under both tests ($t(49) = 3.60$, $p = 0.001$; 39 of 50 queries improved, sign test $p < 0.001$), with a bootstrap 95% confidence interval of $[+0.050, +0.164]$. The effect size is $d = 0.509$, which requires 31 queries at 80% power, so the collection is comfortably adequate for this measure. Recall at rank 10 also improves significantly, from 0.3011 to 0.3502 ($p = 0.042$).

nDCG@10 and P@10 improve by a similar margin and neither reaches significance. Each of the three effects at rank 10 is a little under half the size of the MAP effect, and the two rank-weighted measures remain below the threshold 50 queries can resolve. The pattern is consistent: stemming puts relevant documents into the first ten results that were not there before, which recall detects, while the gain in how those ten are ordered is smaller than this collection can establish.

The gain is uneven. Thirty-nine queries gain 7.13 MAP between them while eleven lose 1.78, three of those accounting for 1.10.

D. A suffix the retrieval evaluation overruled

روغتون is *hospital* and روغتيا is *health*. Both are built on روغ, *healthy*, the first with the place-forming تون and the second with the abstract-noun تيا. The derivation is regular and both suffixes are documented, so an inventory built from the grammar strips them and the query روغتون درملنه becomes درمل روغ.

Whether that conflation costs anything is not obvious, since documents about hospitals naturally use the vocabulary of health. Counting the collection settles it. Of the 1,809 documents containing روغتون, 696 (38%) also contain روغتيا; but of the 5,377 containing روغتيا, only 696 (13%) contain روغتون. The overlap is one-directional. The hospital documents a searcher wants already contain روغتون and are reachable without stemming, so the merge adds none of them; it adds the 4,681 documents that discuss health and never mention a hospital. The judgements agree: of the documents retrieved for this query that contain روغتون, 32 of 43 were judged relevant, while of the 13 containing only روغتيا, none were.

Stripping the suffix also split the word family. Because the pipeline makes a single pass, روغتون lost the suffix while روغتونه spent its pass on the inflectional ه and stopped, so a query for the singular no longer reached a document holding the plural; 430 documents hold an inflected form and no bare روغتون.

The obvious alternative is to run the pipeline twice, and we measured it rather than assuming. A second pass does repair this class of split, but it costs 0.99 points on the development set and 1.81 on held-out text, and it is worse on every retrieval measure as well: MAP falls from 0.6854 to 0.6769, worse on 18 of the 50 queries and better on 8. A second pass removes affixes that the first pass correctly left in place, and that loss outweighs the families it repairs. The pipeline therefore stays at one pass, and the splits that remain are reported in the limitations rather than fixed.

The rule is therefore recorded in the inventory and marked never applied. Gating it is worth 0.0115 MAP and improves all four measures, the only configuration tested that does so (Table IX, lower panel), and it makes all seven surface forms of روغتون reduce to one stem. The cost is 0.22 points of development accuracy and 0.20 of held-out accuracy; on the two external word lists of Section VII-C the same decision raises accuracy. This is the only point where the intrinsic and the retrieval evaluations disagree, and the disagreement is instructive: both analyses of روغتون are defensible for a word in isolation, which is the only way an annotator sees it, so a word-type evaluation cannot detect this class of error in principle.

The similar ستان was tested identically and left in place, because gating it measures slightly worse and افغانستان د ولايتونه falls from 0.1189 to 0.0291: keeping افغانستان whole makes that query match the news agency's own name. روغتون is a common noun whose stripped form collides with a frequent abstract noun, while افغانستان is a proper noun whose stripped form is itself a useful retrieval term.

E. Limitations of the collection

Fifty queries meets the usual TREC-style minimum and is sufficient for the effects on MAP and on recall at rank 10; it is not sufficient for the smaller effects on nDCG@10 and P@10, and we draw conclusions accordingly.

Judging by consensus removes the variation between assessors but does not remove the difficulty of the task, and it leaves no agreement coefficient to report for these labels. An earlier and separate exercise, in which one of the assessors judged three queries independently before the consensus procedure was adopted, gives raw agreement of 63.64% and $\kappa = 0.244$ on 132 documents. That figure describes independent judging, not the labels used here, and disagreement of this magnitude is not unusual in retrieval evaluation: Voorhees [18] showed that substantial variation between assessors leaves the relative ranking of systems largely unchanged. We report it because it shows what the consensus procedure was adopted to avoid.

Passages were formed by grouping consecutive sentences, so one sometimes holds two unrelated news items; the rule adopted was that a passage is relevant if any story in it answers the query. This depresses the absolute scores but affects both runs equally. Finally, stemming conflates word forms and not meanings: كوونكي زده and the Arabic loan محصلين both denote students and share no morphology, which marks the boundary where query expansion would have to take over.

IX. Discussion

A. Correct rules and under-determined rules

A per-rule analysis separates two kinds of failure, and the distinction matters for how the remaining error should be addressed.

Table X gives the precision of the rules that fire most often on the development set: how many times each contributed to the analysis finally selected, and how often that analysis matched the reference. Publishing it is uncomfortable and necessary. A rule can contribute positive points overall and still be wrong half the time it fires, and a reader cannot tell the two apart from an ablation table.

Table X
Precision of the fourteen most frequently firing rules on the development set. "Fires" counts the analyses finally selected in which the rule took part.

Rule	Fires	Correct	Precision
R2 keep short word	398	357	89.7%
R1 drop ه	227	161	70.9%
R1 drop ى	190	149	78.4%
R1 drop ې	160	133	83.1%
−ى	144	129	89.6%
−و	132	109	82.6%
final trim	92	50	54.3%
−ونو	58	54	93.1%
−ه	47	30	63.8%
−ونه	40	36	90.0%
R2 drop ې at 3 letters	39	21	53.8%
R1 drop ی	36	31	86.1%
−ان	36	15	41.7%
R2 drop و at 3 letters	32	18	56.2%

Building that table changed the system. Two rules turned out to fire 31 times between them and be right on none of those occasions: لو, the oblique infinitive, and لې, the feminine past participle. Both bundle an ل into the affix that the reference keeps, either as the infinitive marker (ساتلو → ساتل, not سات) or as part of the root (بریالی → بریال). The rules that remove the final vowel alone already produce the right answer, so these two could only ever take one letter too many. They are now documented and never applied, which is worth 1.07 points on the development set and nothing either way on held-out text. This is the same defect as the standalone verbal ل, which was removed earlier for the same reason: it applied only to nouns where ل belongs to the root and turned كابل into كاب. Two further rules were removed during the work, one taking نى as a unit where the affix is usually the ى alone, and one restoring a feminine ه to any stem that had lost one, which produced كورونو → كورونه.

Four rules remain below 60% precision and are kept. ان is right 15 times in 36, but it is the ordinary animate plural of Pashto and gating it costs 0.40 points on held-out text; its low precision reflects competition with longer analyses rather than a defect in the rule. The two three-letter length rules sit at 53.8% and 56.2% for the reason the next paragraph describes, and the alternative to a coin-flip there is to leave every short inflected word unstemmed.

Other rules are correct but under-determined. The rule removing a final ه from a short word fires 227 times on the development set and is correct 161 of those. لندې → لند is right and خوارې → خوار is wrong, and the two words are indistinguishable in shape. Removing the rule loses more than it saves: overall accuracy falls by 1.44 points. No

Table XI
Accuracy by the affix the reference removes, development set. The grouping is the one in Table III.

What the reference removes	Types	Accuracy
Nominal inflection	1,116	88.8%
Verbal inflection	41	95.1%
Derivation: nouns	79	83.5%
Derivation: adjectives	26	84.6%
No affix, word left whole	1,347	78.4%
Nothing in the inventory	108	29.6%

reformulation conditioned on the surface form can help; only a lexical resource can.

B. What limits the current accuracy

The 511 development-set errors divide into three kinds. 305 of them, 60%, are over-strips, where the system removed more than the reference does; 167, 33%, are under-strips, and 75 of those are words the system left untouched altogether; the remaining 39 are the same length as the reference but differ in which letters survive. The held-out set behaves the same way, 54% and 41%. Over-stripping is therefore the larger problem on both sets, which is what one would expect of a system whose scorer rewards removing affix material.

Table XI locates the errors. Each type is assigned to the group of the rule that explains the reference's removal, under the same grouping as Table III, so that the two tables can be read against each other. Accuracy is high wherever the reference removes an affix the inventory holds: 88.8% on nominal inflection, 83.5% on noun-forming derivation, 95.1% on verbal inflection. The two weak rows are the ones that do not depend on the inventory being right. 108 types have a reference the inventory cannot explain at all, and we reach 29.6% on those. The larger group is the 1,347 types the reference leaves whole, where we are right 78.4% of the time: 291 words stripped that should not have been, the biggest single source of error in the system.

We had attributed that bucket to loanwords, and the data does not support it. Pashto marks native vocabulary orthographically: a word containing one of ک ن څ خ ښ ږ ړ ډ ټ ې ۍ is certainly not a borrowing, though a word without one may be either, so the test is one-sided. Among the types the reference leaves whole we are wrong on 28.3% of the certainly-native ones and 18.8% of the rest. Native words are the harder half, not the easier one, and the errors bear it out: خواړه, وړاندې, داسې, پاتې, یوازې and پورې are ordinary Pashto words ending in ه or ې that the length rules cut. The shortfall is the ې/ی/ه ambiguity, and it falls on native vocabulary rather than on borrowings.

We tested the obvious remedy and report the result because it is negative. A 257-entry list of development-set words that should be left unchanged raises development accuracy by 8.5 points and held-out accuracy by zero: of the 51 held-out types needing the same treatment, none appear in the list. The list is pure memorisation of one sample. The generalizing form of the same idea is a lexicon of words that must not be stripped. On the evidence above it would need to cover ordinary native vocabulary rather than borrowings alone, which makes it a larger undertaking than we first assumed, and we leave it to future work.

X. Limitations

The retrieval evaluation of Section VIII rests on 50 queries, which is enough to establish the effects on MAP and on recall at rank 10 but not the smaller effects on nDCG@10 and P@10. Its judgements are adjudicated rather than independent, which removes the variation between assessors at the cost of leaving no agreement coefficient for the labels themselves. The document collection was scraped from BBC Pashto, VOA Pashto, Pajhwok Afghan News, Bakhtar News Agency, Shamshad News, TOLOnews, Afghanistan International and *Nan Taki Asia*. The proportion taken from each was not recorded, nor was the date range, and the passages are groups of consecutive sentences rather than whole articles. The collection also shares two outlets with the corpus the lexicon was drawn from, so the two overlap to an unknown degree; this affects both runs equally, since they index the same documents and differ only in whether the stemmer is applied.

Of 118 affix rules, 112 cite a published grammar by chapter or section and the six that do not are entries the engine never applies. Of 38 verbs in the dictionary, 23 are attributed to a printed grammar and 15 were checked by the author, a native speaker of Pashto, against his own judgement rather than a published paradigm. The source of each is recorded in Appendix B, so the two groups can be told apart.

Compounds are kept whole, which is a policy choice rather than a solution: it avoids a class of error at the cost of never conflating لاسلیک with لاس. The annotation does not follow the policy perfectly either, and an audit during development found types where the reference itself splits a compound. The released audit script reports every removal that matches no documented affix, which is where such cases surface.

The corpus is drawn from news and general text, so the vocabulary of specialised domains is under-represented.

XI. Conclusion

We presented a rule-based stemmer for Pashto built from an affix inventory of 118 rules taken from published grammars, each recording its source, its productivity and an explicit decision about whether it may be applied. No machine learning is used, and every decision the system makes can be traced to a rule or a dictionary entry.

On 499 held-out word types from news text, annotated blind and never used during development, the system reaches 72.55% exact-match accuracy against 40.48% for

the published baseline, and 72.5% against 16.9% on the types that require a strip. On a 2,717-type development set it reaches 80.97% against 50.68%, with Paice's understemming index falling from 0.919 to 0.296.

On a retrieval collection of 50 queries over 44,814 news documents with 1,891 relevance judgements adjudicated by three native speakers, stemming raises mean average precision from 0.5785 to 0.6854 ($p = 0.001$; 39 of 50 queries improved, sign test $p < 0.001$) and recall at rank 10 from 0.3011 to 0.3502 ($p = 0.042$). The improvement in nDCG@10 and P@10 is positive and not significant at this number of queries. One suffix, the place-forming تون, is recorded in the inventory and never applied, because removing it merges a place with the quality it is named for and costs more in retrieval than its 0.22 points of word-type accuracy are worth.

Three observations generalise beyond this particular system. Exact-match accuracy is not a sufficient measure for stemming in a language where a large share of word types carry no affix, because it rewards inactivity; reporting the two halves of the task separately costs nothing and prevents the error. An evaluation set drawn from a single orthographic convention will conceal defects that a second source exposes immediately, as the *ye* variants did here. And when a rule-based system finds no affix, the useful response is to ask what the inventory is missing rather than to remove characters regardless: the one rule in this work that guessed a boundary instead of finding one was measured against three references, lost on both careful annotations, and was removed.

The stemmer, the affix inventory with its sources, the verb dictionary, both annotated sets and the scripts that reproduce every figure in this paper are released under an MIT licence at https://github.com/Khairullah-Ibrahim-Khail/Pashto_Stemmer. Every table and every figure in this paper is reproduced by the released scripts.

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Appendix

The 118 rules in full, in the order the engine holds them. "Type" gives the side and whether the rule is inflectional or derivational; "min" is the shortest stem the rule is allowed to leave; "strip" is whether the engine may remove the affix at all; "source" is what backs the rule: a published grammar, this project's inventory document, and `gated` for an entry that is never applied and appeals to no grammar. The entries marked *no* under "strip" are documented and never applied, for the reasons given in Section IV.B. This table is generated directly from the rule objects, so it cannot drift from the implementation.

Affix	Type	min	POS	Productivity	Confidence	Strip	Source
Nominal inflection							
تانه	suf, infl	3	N	prod.	high	yes	W
تانو	suf, infl	3	N	prod.	high	yes	W
ګانو	suf, infl	2	N	prod.	high	yes	T&R M4/F3
ګاني	suf, infl	2	N	prod.	high	yes	T&R F2/F3
ګونه	suf, infl	2	N	prod.	high	yes	T&R ch.5 collective numbers
ګونو	suf, infl	2	N	prod.	high	yes	T&R ch.4 noun inflection
اتو	suf, infl	3	N	prod.	high	yes	T&R ch.4 loanword case paradigm
انو	suf, infl	2	N	prod.	high	yes	T&R M1
اني	suf, infl	3	N	prod.	high	yes	T&R F2
ڇات	suf, infl	3	N	prod.	high	yes	T&R ch.2 Persian nominal suffixes
ونه	suf, infl	2	N	prod.	high	yes	T&R M2
ونو	suf, infl	2	N	prod.	high	yes	T&R M2
ګان	suf, infl	2	N	prod.	high	yes	T&R ch.4 animate plurals
ات	suf, infl	3	N	prod.	high	yes	T&R ch.2 Arabic borrowings
ان	suf, infl	3	N	prod.	high	yes	T&R M1
ون	suf, infl	3	N	prod.	high	yes	N&K
وو	suf, infl	3	N	prod.	high	yes	T&R
وې	suf, infl	3	N	prod.	high	yes	T&R
یو	suf, infl	3	N	prod.	high	yes	T&R F2/F3
ه	suf, infl	3	N/ADJ	prod.	high	yes	T&R ch.4 gender, ch.5 agreement
و	suf, infl	3	N/ADJ	prod.	high	yes	T&R M3/Adj2
ی	suf, infl	3	N/ADJ	prod.	high	yes	T&R ch.4 feminine noun classes
ې	suf, infl	2	N/ADJ	prod.	high	yes	T&R F1/Adj
Verbal inflection							
ېدا	suf, infl	2	V	prod.	high	yes	R&T 13.16
ېدل	suf, infl	2	V	prod.	high	yes	T&R ch.6 derivative verbs
ېده	suf, infl	2	V	prod.	high	yes	T&R ch.6 imperfective past stems
ېدو	suf, infl	2	V	prod.	high	yes	R&T 13.16
ېږم	suf, infl	2	V	prod.	high	yes	R&T 13.16
ېږو	suf, infl	2	V	prod.	high	yes	R&T 13.16
ېږی	suf, infl	2	V	prod.	high	yes	R&T 13.16
لو	suf, infl	2	V	prod.	high	no	gated
ول	suf, infl	2	V	prod.	high	yes	T&R ch.6 transitive derived verbs
ېد	suf, infl	2	V	prod.	high	yes	R&T 13.16
ېږ	suf, infl	2	V	prod.	high	yes	R&T 13.16
ئ	suf, infl	3	V	prod.	high	yes	T&R ch.6 imperative paradigm
ل	suf, infl	2	V	prod.	low	no	gated
Derivation: nouns							
یدونکي	suf, der	3	N/ADJ	prod.	high	yes	R&T 13.16
ېدونکي	suf, der	3	N/ADJ	prod.	high	yes	R&T 13.16
خانه	suf, der	3	N	prod.	high	yes	T&R ch.4 derived nouns of place
داني	suf, der	3	N	borr.	med.	yes	Penzl §51 containers
ستان	suf, der	3	N	borr.	med.	no	inv §2.3
غالی	suf, der	3	N	lex.	low	no	T&R ch.4 nouns of place
والو	suf, der	3	N	prod.	high	yes	T&R ch.4 oblique plural formations
والی	suf, der	3	N	prod.	high	yes	N&K
ونکو	suf, der	3	N	prod.	high	yes	T&R ch.6 participle case shifts
ونکی	suf, der	3	N/ADJ	prod.	high	yes	T&R
ونکي	suf, der	3	N/ADJ	prod.	high	yes	T&R ch.6 active participle paradigm
کوټی	suf, der	3	N	restr.	low	no	Penzl §53 diminutives
ګلوي	suf, der	3	N	restr.	high	yes	T&R ch.4 abstract noun formations
ګوټی	suf, der	3	N	restr.	low	no	Penzl §53 diminutives
اوی	suf, der	3	N	restr.	med.	no	T&R ch.6 verbal nouns
بان	suf, der	3	N	borr.	high	yes	Penzl §52 suffixes of the doer
توب	suf, der	3	N	prod.	high	yes	N&K
تون	suf, der	3	N	prod.	high	no	N&K

Affix	Type	min	POS	Productivity	Confidence	Strip	Source
تیا	suf, der	2	N	prod.	high	yes	T&R ch.4 noun derivation
خور	suf, der	3	N	borr.	med.	yes	T&R ch.4 synthetic compounds
دار	suf, der	3	N	prod.	high	yes	T&R ch.2 Persianate compounds
دان	suf, der	4	N	borr.	med.	yes	Penzl §51 containers
زار	suf, der	3	N	borr.	med.	yes	T&R ch.4 locational suffixes
لیک	suf, der	3	N	prod.	med.	no	T&R ch.4 administrative compounds
نکو	suf, der	3	N/ADJ	prod.	high	yes	T&R ch.6 participle variations
نکی	suf, der	3	N/ADJ	prod.	high	yes	T&R ch.6 active participle paradigm
نکې	suf, der	3	N/ADJ	prod.	high	yes	T&R ch.6 active participle paradigm
وال	suf, der	3	N	restr.	med.	no	T&R ch.4 suffixes of affiliation
وان	suf, der	3	N	restr.	low	no	Penzl §52 suffixes of the doer
ولې	suf, der	3	N	restr.	low	no	T&R ch.4 native abstract derivations
وتی	suf, der	3	N	prod.	high	yes	Khan23
وکی	suf, der	3	N	lex.	low	no	Penzl §53 diminutives
پال	suf, der	3	N/ADJ	prod.	high	yes	T&R ch.4 synthetic compound stems
پوه	suf, der	3	N	prod.	high	yes	T&R ch.4 compound nouns
ځای	suf, der	3	N	prod.	high	yes	T&R ch.4 derived nouns of place
ځیو	suf, der	3	N	prod.	high	yes	W
ښنه	suf, der	3	N	restr.	high	yes	T&R ch.4 action nouns from verbs
کار	suf, der	3	N	prod.	high	yes	Penzl §52 suffixes of the doer
ګار	suf, der	3	N	borr.	high	yes	Penzl §52 suffixes of the doer
ګره	suf, der	3	N	prod.	high	yes	N&K
ګړی	suf, der	3	N	prod.	high	yes	Penzl §52 suffixes of the doer
یال	suf, der	3	N	restr.	med.	no	T&R ch.5 suffixal adjectives
ار	suf, der	4	N	restr.	med.	no	T&R ch.4 noun derivations
اک	suf, der	4	N	restr.	high	yes	Penzl §53 diminutives
نه	suf, der	3	N	prod.	high	yes	W
نې	suf, der	3	N	prod.	med.	no	gated
ځی	suf, der	3	N	prod.	high	yes	W
چې	suf, der	3	N	lex.	low	no	Penzl §52 suffixes of the doer
ښت	suf, der	3	N	prod.	high	yes	W
ګر	suf, der	3	N	prod.	high	yes	Penzl §52 suffixes of the doer
ګی	suf, der	3	N	prod.	high	yes	Khan23
یت	suf, der	3	N	prod.	high	yes	T&R ch.2 morphological loan layers
یې	suf, der	3	ADJ/N	prod.	high	yes	T&R ch.5 noun-to-adjective suffixes
ا	suf, der	4	N	restr.	low	no	T&R ch.4 abstract suffixes
ت	suf, der	4	N	restr.	med.	no	T&R ch.2 loan morphology
ک	suf, der	4	N	restr.	med.	no	T&R ch.3 consonant alternations
ی	suf, der	3	N/ADJ	prod.	high	yes	W
Derivation: adjectives							
انه	suf, der	3	ADJ	prod.	high	yes	T&R ch.5 suffixal derivations
مند	suf, der	3	ADJ	prod.	high	yes	T&R ch.5 suffixal adjectives
ناک	suf, der	3	ADJ	prod.	high	yes	T&R ch.5 adjective derivation
چکی	suf, der	3	ADJ	prod.	high	yes	Khan23
یزه	suf, der	3	ADJ	prod.	high	yes	T&R ch.5 modern formations
یزو	suf, der	3	ADJ	prod.	high	yes	T&R ch.5 oblique adjectival agreement
جن	suf, der	3	ADJ	restr.	high	yes	T&R ch.5 noun-to-adjective suffixes
من	suf, der	3	ADJ	prod.	high	yes	T&R ch.5 regional suffixal variants
نی	suf, der	3	ADJ	prod.	med.	no	gated
نۍ	suf, der	3	ADJ	prod.	med.	no	gated
ور	suf, der	3	ADJ	restr.	med.	no	T&R ch.5 suffixal adjectives
وړ	suf, der	3	ADJ	prod.	high	yes	W
ین	suf, der	3	ADJ	restr.	med.	no	T&R ch.5 suffixal adjectives
یز	suf, der	3	ADJ	prod.	high	yes	T&R ch.5 modern formations
Derivation: verbs							
لې	suf, der	3	ADJ/V	prod.	high	no	T&R
Prefixes							
بلا	pre, der	3	ADJ	borr.	med.	yes	T&R ch.5 intensive modifiers
بیا	pre, der	3	V/N	prod.	high	yes	T&R ch.7 adverbs
غیر	pre, der	3	ADJ	borr.	high	no	T&R ch.2 morphological borrowings
نیم	pre, der	3	ADJ/N	restr.	high	yes	T&R ch.5 numbers and modifiers
بې	pre, der	3	ADJ/N	prod.	high	no	T&R ch.5 prefixed adjectives
سر	pre, der	4	N	restr.	med.	no	T&R ch.4 nominal compounding
ضد	pre, der	3	N	borr.	med.	yes	T&R ch.2 modern loan vocabulary
لا	pre, der	3	any	borr.	med.	yes	T&R ch.2 Arabic/Persian bound morphemes
نا	pre, der	3	ADJ/N	prod.	high	no	T&R ch.5 prefixed adjectives
هم	pre, der	3	any	prod.	high	yes	T&R ch.2 Persian compounds

The 38 strong verbs the dictionary holds, with the 315 surface forms they map. These are the verbs whose present stem cannot be derived from the infinitive. The last column names the source of each: a published grammar, or the author's own judgement as a native speaker where no published paradigm was consulted. The dictionary returns a lemma, so it is off by default and belongs to the lemmatization mode described in Section IV-G. This table is generated from the dictionary itself, so it cannot claim a verb the software does not hold.

Infinitive	Present stem	Past	Forms	Gloss, source
کول	کو	کاوه / کړ	23	to do/make, Robson & Tegey
کېدل	کیږ / کېږ	شول / شو	18	to become/happen, Robson & Tegey
تلل	ځ	تل / لاړ	16	to go, Robson & Tegey
راتلل	راځ	راغی / راغل	12	to come, Robson & Tegey
وړل	وړ	یور / یووړ	11	to carry/take away, Robson & Tegey
راوړل	راوړ	راووړ	6	to bring, Robson & Tegey
خوړل	خور	خوړ	9	to eat, Tegey & Robson (1996)
څښل	none	none	6	to drink, author
لیدل	وین	لید	10	to see, Tegey & Robson (1996)
کتل	ګور	کت / کوت	8	to look, Tegey & Robson (1996)
اخیستل	اخل	اخیست	9	to take/buy, Tegey & Robson (1996)
نیول	نیس	نیو	8	to catch/hold, Tegey & Robson (1996)
ویل	وای	ویل	10	to say, Tegey & Robson (1996)
ورکول	ورکو	ورکړ	9	to give, Robson & Tegey
غوښتل	غواړ	غوښت	10	to want, Tegey & Robson (1996)
موندل	موم	موند	8	to find, Tegey & Robson (1996)
لوستل	لول	لوست	6	to read, Tegey & Robson (1996)
لرل	لر	لرل / درلود	11	to have, author
وژل	وژن	ووژه / وژل	7	to kill, Tegey & Robson (1996)
اورېدل	اور	اورېد	8	to hear, author
پېژندل	پېژن / پېژن	پېژند / پېژند	7	to know/recognize, Tegey & Robson (1996)
رسېدل	none	none	9	to arrive / reach, author
پوهېدل	none	none	7	to understand, author
اوسېدل	none	none	6	to live / reside, author
ښودل	ښی	ښود	7	to show, Tegey & Robson (1996)
کېښودل	زد	ایښود / کېښود	7	to put/place, author
پرېښودل	پرېزد	پرېښود	5	to leave/abandon, author
الوتل	الوز	الوت	6	to fly, Tegey & Robson (1996)
ختل	خېژ	خت / خوت	6	to climb/rise, Tegey & Robson (1996)
پرېوتل	پرېوځ	پرېوت	5	to fall, author
کښېنستل	کښېن	ناست / کښېناست	6	to sit down, author
پاڅېدل	none	none	5	to stand up / rise, author
ګڼل	ګڼ	ګڼل / ګانه	6	to count/consider, author
بلل	بول	بلل	6	to call/consider, Tegey & Robson (1996)
سپارل	none	none	6	to hand over / entrust, author
ژغورل	none	none	5	to save / rescue, author
ایستل	باس	وېست / ایست	6	to pull out/extract, Tegey & Robson (1996)
کارول	none	none	5	to use / operate, author

A further 12 strong verbs were identified during the work and are *not* in the dictionary. They are listed because a reader extending the dictionary should start here, and so that the main table is not read as covering them.

Infinitive	Present stem	Past	Gloss	Source
اغوستل	اغوند	اغوست	to wear	Tegey & Robson (1996)
اوړل	اوړ	اوښت / وښت	to pass/cross	Tegey & Robson (1996)
چاودل	چو	چاود	to explode	Tegey & Robson (1996)
نغښتل	نغاړ	نغښت	to wrap/roll up	Tegey & Robson (1996)
ویشتل	ول	وېشت	to shoot	Tegey & Robson (1996)
وتل	وځ	وت	to go out	unverified
رودل	رو	رود	to suck	Tegey & Robson (1996)
ورتلل	ورځ	ورغل / ورتل	to go there	unverified
درتلل	درځ	درغل / درتل	to go to you	unverified
راکول	راکو	راکړ	to give here	unverified
درکول	درکو	درکړ	to give to you	unverified
پرانیستل	پرانیز	پرانیست	to open	unverified